

Examiner
only

- (c) lestyn claims that changing the shape of the coat hanger from Shape 1 to Shape 2 in 16 ms in the magnetic field will increase its temperature by less than 1 °C. Determine, using appropriate calculations, whether or not lestyn is correct. (Density of aluminium = 2 700 kg m⁻³, specific heat capacity of aluminium = 897 J kg⁻¹ K⁻¹.) [5]

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- (d) For medical research, it is decided to investigate the effect of this strong magnetic field (2.1 T) on patients with metal replacement joints to see if the metal joints become hot or undergo large forces (during MRI scans). Discuss the ethics of such an experiment. [3]

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